AUM
American University Of The Middle East

Name:	
ID:	
Date:	

Bachelor of Science in Mechanical Engineering
Worksheet

Mechanical Engineering (ME) Requirements (at least 62 credit hours)

Mechanical Sciences (12 credit hours)

	Course Code	Course Title	CR	Pre-Req, ConP	Contact Hours	Other Information	Semester	Comments
☐	ME 270	Basic Mechanics I	3	Pre: PHYS 172 (min. grade C-), MA 166 (min. grade C-); ConP: MA 261, ENGR 132	Lect.: 3h			
☐	ME 274	Basic Mechanics II	3	Pre: ME 270, ENGR 132 (C-) ConP: MA 262	Lect.: 3h			
☐	ME 323	Mechanics Of Materials	3	Pre: ME 270	Lect.: 3h			
☐	MSE 230	Structure and Properties of Materials	3	Pre: CHM 115 (min. grade C-), MA 165 (min grade C-)	Lect.: 3h			

Mechanical Engineering Seminars (1 credit hour)

	Course Code	Course Title	CR	Pre-Req, ConP	Contact Hours	Other Information	Semester	Comments
☐	ME 290	Global Engineering Professional Seminar	1	Pre: COM 114, ENGL 106	Lect.: 1h			

Systems, Measurements and Control (10 credit hours)

	Course Code	Course Title	CR	Pre-Req, ConP	Contact Hours	Other Information	Semester	Comments
☐	EE 201	Linear Circuit Analysis I	3	Pre: ENGR 131, PHYS 172, MA 166 (min grade C-) ConP: MA 261	Lect.: 3h			

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☐	EE 207 (L)	Electronic Measurement Techniques Lab	1	ConP: EE 201	Lab.: 3h			
☐	ME 365 (L)	Systems and Measurements	3	Pre: EE 201, ME 274, MA 262, EE 207	Lect.: 2h; Lab: 2h			
☐	ME 375	System Modeling and Analysis	3	Pre: ME 365, MA 303	Lect.: 3h			

(ME) Mechanical Sciences Credits Planned: _______
(ME) Seminars Credits Planned:________
(ME) Systems, Measurements and Control Planned:______
Subtotal Credits Planned: ________

(ME) Mech. Sciences Credits Completed _____
(ME) Seminars Credits Completed:________
(ME) Systems, Measurements and Control Completed:______
Subtotal Credits Completed: ________

ME) Mech. Sciences Cr. Remaining: _____
(ME) Seminars Credits Remaining:________
(ME) Systems, Measurements and Control Remaining:______
Subtotal Credits Completed: ________

Thermal / Fluid Sciences (11 credit hours)

	Course Code	Course Title	CR	Pre-Req, ConP	Contact Hours	Other Information	Semester	Comments
☐	ME 200	Thermodynamics I	3	Pre: CHM 115 ConP: MA 261, ENGR 132	Lect.: 3h			
☐	ME 309 (L)	Fluid Mechanics	4	Pre: ME 263, ME 274, MA 262	Lect.:3h; Lab: 2h			
☐	ME 315 (L)	Heat and Mass Transfer	4	Pre: ME 309, ME 365, MA 303	Lect.:3h; Lab: 2h			

Design Courses (10 credit hours)

	Course Code	Course Title	CR	Pre-Req, ConP	Contact Hours	Other Information	Semester	Comments
☐	ME 263 (L)	Introduction to Mechanical Engineering Design, Innovation and Entrepreneurship	3	Pre: ME 270, COM 114, ENGL 106, ENGR 132 ConP: ME200, MA 262, ME 290, CGT 163	Lect.:2h; Lab: 2h			
☐	ME 352 (L)	Machine Design I	4	Pre: ME 263, ME 274 and ME 323	Lect.:3h; Lab: 2h			
☐	ME 463 (L)	Engineering Design	3	Pre: ME 352, MSE 230 ConP: ME 315, ME 375	Lect.:1; Lab: 2h			

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Restricted Electives (6 credit hours)

Complete two of the following three courses. The remaining course may be taken as a professional elective or as a free elective: ME 300, ME 452, ME 475 (L)

	Course Code	Course Title	CR	Pre-Req, ConP	Contact Hours	Other Information	Semester	Comments
☐	ME 300	Thermodynamics II	3	Pre: ME 200, ME 263	Lect.: 3h			
☐	ME 452	Machine Design II	3	Pre: ME 352, MSE 230	Lect.: 3h			
☐	ME 475 (L)	Automatic Control Systems	3	Pre: ME 375	Lect.: 2h; Lab: 2h			

Professional Electives (12 credit hours)

Select 4 professional elective courses from the pool of Professional Electives in Mechanical Engineering:

	Course Code	Course Title	CR	Pre-Req, ConP	Contact Hours	Other Information	Semester	Comments
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	Course Code	Course Title	CR	Pre-Req, ConP	Contact Hours	Other Information	Semester	Comments
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Thermal / Fluid Sciences Credits Planned (min):_________
Design Courses Credits Planned (min)
Restricted Electives Credits Planned (min): _________
Professional Electives Planned (min): ________
Subtotal Credits Planned (min): _________

Thermal / Fluid Sciences Credits Completed (min):_________
Design Courses Credits Completed (min)
Restricted Electives Credits Completed (min): _________
Professional Electives Planned (min): _______
Subtotal Credits Completed (min): _________

Thermal/Fluid Sci. Cr. Remaining (min):_____
Design Courses Cr. Remaining (min)_____
Restricted Elec. Cr. Remaining (min): __
Professional Elec. Planned (min): __
Subtotal Credits Remaining (min): _________

General Engineering Requirements (6 credit hours)

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Introduction to Engineering (4 credit hours):

	Course Code	Course Title	CR	Pre-Req, ConP	Contact Hours	Other Information	Semester	Comments
☐	ENGR 131	Transforming Ideas to Innovation I	2	-	Lect.: 2h; Lab.: 2h			
☐	ENGR 132	Transforming Ideas to Innovation II	2	Pre: ENGR 131	Lect.: 2h; Lab.: 2h			
OR								
☐	ENGR 100	First-Year Engineering Lectures	1	-	Lect.: 1h			
☐	ENGR 126	Engineering Problem Solving and Computer Tools	3	-	Lect.: 3h			

General Engineering Requirement for Mechanical Engineering (2 credit hours):

	Course Code	Course Title	CR	Pre-Req, ConP	Contact Hours	Other Information	Semester	Comments
☐	CGT 163	Graphical Communication and Spatial Analysis	2	–	Lect.: 2h; Lab:2h			

General Engineering Credits Planned: ______
General Engineering Credits Completed: ______
General Engineering Credits Remaining: ______

Mathematics Requirement (19 credit hour):

	Course Code	Course Title	CR	Pre-Req, ConP	Contact Hours	Other Information	Semester	Comments
☐	MA 165	Analytic Geometry and Calculus I	4	Pre: MAT 110 or MA 158 or passing Math Placement Test	Lect.: 3h; Lab.: 2h			
☐	MA 166	Analytic Geometry and Calculus II	4	Pre: MA 165	Lect.: 4h			
☐	MA 261	Multivariate Calculus	4	Pre: MA 166	Lect.: 4h			

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☐	MA 262	Linear Algebra and Differential Equations	4	Pre: MA 261	Lect.: 4h			
☐	MA 303	Differential Equations and Partial Differential Equations for Engineering and the Sciences	3	Pre: MA 262 OR (MA 265 and MA 266)	Lect.: 3h			

Mathematics Credits Planned: ______
Mathematics Credits Completed: ______
Mathematics Credits Remaining: ______

Science Requirements (14-15 credit hours):

	Course Code	Course Title	CR	Pre-Req, ConP	Contact Hours	Other Information	Semester	Comments
☐	PHYS 172	Modern Mechanics	4	ConP: MA 165	Lect.: 3h; Lab.: 2h			
☐	PHYS 241	Electricity and Optics	3	Pre: PHYS 172 ConP: MA 166	Lect.: 3h			
☐	CHM 115	General Chemistry I	4	ConP: MA 165	Lect.: 3h; Lab.: 2h			

Science Selectives (3-4 credit hours)

	Course Code	Course Title	CR	Pre-Req, ConP	Contact Hours	Other Information	Semester	Comments
☐	CHM 112	General Chemistry	3	Pre: CHM 115	Lect.: 3h			
☐	CHM 116	General Chemistry II	4	Pre: CHM 115	Lect.: 3h; Lab.: 2h			
☐	BIOL 126	Human Biology	3	-	Lect.: 3h			
☐	PHYS 322	Intermediate Optics	3	Pre: PHYS 272 or PHYS 241	Lect.: 3h			
☐	PHYS 342	Modern Physics	3	Pre: PHYS 272 or PHYS 241	Lect.: 3h			
☐	BIOL 110	Fundamentals of Biology I	4	-	Lect.:3h; Lab.: 2h			

☐	(BIOL 121	Biology I: Diversity, Ecology, and Behavior	2	-	Lect.:2			
☐	and BIOL 135)	First year Biology Laboratory	2	ConP: BIOL 121, CHM 115	Lab.: 3			
☐	EAPS 125	Environmental Science and Conservation	3	-	Lect.: 3h			

Science Credits Planned: ______

Science Credits Completed: ______

Science Credits Remaining: ______

Liberal Arts Requirements (27-28 credits)

Refer to the Liberal Arts Department Course Catalogue

English Language and Communication Skills (10 credits).

	Course Code	Course Title	CR	Pre-Req, ConP	Contact Hours	Other Information	Semester	Comments
☐	ENGL 100	English for Academic Studies	3	-	Lect.: 3h			
☐	ENGL 106	First-Year Composition	4	Pre: ENGL 100	Lect.: 4h			
☐	COM 114	Fundamentals Of Speech Communication	3	Pre: ENGL 100	Lect.: 3h			

General Education Electives (17-18 credits):

	Course Code	Course Title	CR	Pre-Req, ConP	Contact Hours	Other Information	Semester	Comments
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Liberal Arts Credits Planned: ______

Liberal Arts Credits Completed: ______

Liberal Arts Credits Remaining: ______

Free Electives (3 credit hours):

	Course Code	Course Title	CR	Pre-Req, ConP	Contact Hours	Other Information	Term	Comments
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* Remedial courses are courses that must be taken by student in order to build up certain academic knowledge or skills like math or English.

Minimum Total Credits Required for Degree: 132

Total Credits Planned: _________

Total Credits Completed: _________

Total Credits Remaining: _________